



Enhance four-dimensional cone-beam computed tomography (4D-CBCT) from sparse view acquisitions using a novel deep learning model

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ABSTRACT

Purpose: Four-dimensional cone-beam computed tomography (4D-CBCT) enables time-resolved imaging of moving targets but requires long scan times and high radiation doses for sufficient projections per respiratory phase. This study proposes a deep learning-based projection interpolation model with data-aware convolutional design to enable sparse-view 4D-CBCT with reduced imaging time and dose.

Methods: We developed a two-stage workflow, Interpolation + MoCo, to enhance sparse-view 4D-CBCT. The Interpolation stage employs dynamic region-aware and rotation-aware convolutional modules to predict missing projections by adjusting weights based on angular distance and spatial subregions. The MoCo stage performs motion compensation using a deep learning-based deformable image registration (DIR) model. Both models were trained and evaluated on 100 full-view scans from 20 lung cancer patients. Full-view data (~2500 projections) were downsampled to 900 sparse projections, with remaining projections used as targets. The interpolation model was trained on 90,190 sample sets (one set = {two sparse projections, one target}), validated on 30,245, and tested on 30,084. For DIR, reconstructed volumes from interpolated and original sparse projections formed phase pairs (training: 5400; validation: 200; test: 200). Performance was assessed using root mean square error (RMSE), structural similarity index measure (SSIM), and peak signal-to-noise ratio (PSNR) against motion-compensated full-view reconstructions.

Results: The workflow enables high-quality 4D-CBCT from 1-minute sparse acquisitions. Interpolation achieved SSIM: 0.9657 ± 0.0155 , RMSE: $(8.38 \pm 0.98) \times 10^{-4} \text{ mm}^{-1}$, PSNR: $28.44 \pm 1.56 \text{ dB}$. After MoCo, results improved to SSIM: 0.9905 ± 0.0061 , RMSE: $(32.46 \pm 5.14) \times 10^{-4} \text{ mm}^{-1}$, PSNR: $35.09 \pm 1.65 \text{ dB}$.

Conclusions: Interpolation + MoCo enables accurate, motion-compensated 4D-CBCT reconstruction from sparse-view data, supporting faster, lower-dose imaging for radiotherapy motion management.

1. Introduction

In radiotherapy, four-dimensional cone-beam computed tomography (4D-CBCT) is a valuable imaging technique that enables time-resolved target localization [1–4]. However, respiratory phase sorting [5], which is commonly used to reduce motion artifacts, often leads to a limited number of projections per phase, significantly degrading image quality. To mitigate the resulting under-sampling issue, a typical 4D-CBCT scan requires 3–4 min [6–8] for full-fan scans and 8–10 min [9] for half-fan scans, which is considerably longer than the widely used three-dimensional (3D) CBCT (approximately 1 min) [4], thereby limiting its clinical utilities.

Due to its great success in medical image processing [10–13], deep learning has been increasingly adopted to enhance 4D-CBCT image

quality while enabling reduced imaging dose and scanning time.

Estimating missing projections from known ones is a natural and intuitive solution. SynCNN [10] directly inferred the missing projections from the available ones. RFI-Net [14] synthesized missing projections from adjacent ones in the projection feature domain. Lu et al. [15] proposed two deep residual U-Net (DRU) models to first optimize linearly interpolated projections from sparse views, and subsequently enhance the CBCT image reconstructed using the Feldkamp-Davis-Kress (FDK) algorithm [16]. However, these models fail to account for the varying contributions of known projections, which are influenced by their angular distances from the missing projection.

Image domain denoising aims to enhance image quality directly from the corrupted CBCT images. The first approach aims to compensate for missing data using information from other images. Jiang et al. [17]

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addressed the missing information in target phase from other phases in the image feature domain using a deformable convolutional network. DeCNN [18], an abbreviation for dual-encoder convolutional neural network, compensates for the under-sampled target-phase features from an average 4D image. MeCNN [19] (merging-encoder convolutional neural network) merges the image features from prior CT images and under-sampled CBCT images. Subsequently, these concatenated features were fed to a decoder for a high-quality reconstruction. The second approach focuses on learning the mapping from under-sampled data to full-view quality. Zhang et al. [20,21] employed a 3D U-Net with residual learning to mitigate undersampling-induced streaking artifacts in 4D-CBCT. Based on the enhanced 4D-CBCT images, they estimated the inter-phase deformation vector field (DVF) to compensate for the missing projections in the target phase using information from other phases. Jiang et al. [22] proposed a symmetric residual convolutional neural network (SR-CNN) designed to learn the restoration patterns from under-sampled reconstructions. The performance of these models is limited by the artifact characteristics learned from the training data, and consequently degrades under complex and diverse noise scenarios [23].

In this work, we propose an Interpolation + MoCo workflow to improve the quality of sparse-view 4D-CBCT, as illustrated in Fig. 1. The term “Interpolation”¹ refers to a deep learning-based interpolation model that predicts missing projections from sparse-view acquisitions, and then generates full-view projections. “MoCo”, short for motion compensation, is the second stage of the workflow, involving a motion-compensated FDK reconstruction [24,25] to mitigate respiration-induced undersampling artifacts. The contributions of this work are as follows:

(1) We propose two customized convolutional modules named Dynamic Region- and Rotation-aware Convolution (DRRConv) and Dynamic Region- and Rotation-aware Transposed Convolution (DRRTransConv) in the interpolation model. They adaptively adjust weights, enabling efficient prediction of missing projections by accounting for differences in angular distances and spatial sub-regions.

(2) The proposed MoCo incorporates a deep learning-based deformable image registration (DIR) model for automation, and sets the DVF to zero outside the thorax to improve image quality in regions unaffected by respiratory motion, such as muscles.

The rest of this paper is organized as follows. Section 2 introduces our model’s structure and experiment settings. The model was trained and evaluated using real patient data. The results are presented in section 3 and discussed in section 4. Section 5 concludes our work.

2. Materials and methods

2.1. Method overview

Fig. 2 shows a schematic representation of our work². The training framework is composed of the following blocks: (a) an interpolation model that predicts a new projection according to its two adjacent projections. By gathering the interpolated and sparse-view projections, we adopted FDK algorithm to reconstruct an initial 4D-CBCT; (b) a DIR model that estimates a DVF to map images from a source respiratory phase to a target phase. During testing, the DIR model is combined with an intrathoracic mask module to generate an intrathoracic DVF for motion compensation. It aims to compensate for the missing target-

phase data by deforming data from other phases³. In the following subsections, we detail the two blocks.

The test framework includes: (a) evaluation on Interpolation—This compares the volume reconstructed using a combination of sparse-view and interpolated projections against the corresponding full-view reconstruction; (b) evaluation on Interpolation + MoCo—This assesses the effectiveness of applying motion compensation to the interpolated volume by comparing it with the corresponding motion-compensated full-view volume.

2.2. Network architectures

2.2.1. Interpolation model structure

As illustrated in Fig. 3(a), the interpolation model adopts a U-Net backbone [26] integrated with our DRRConv and DRRTransConv. In the interpolation model, we employed the layer normalization [27] to learn consistent feature representations, regardless of the input’s distribution. Given two adjacent projections ($P_{\alpha1}$ and $P_{\alpha2}$) and a target interpolated angle β , the interpolation model adaptively adjusts its weights according to the angular distances ($\beta-\alpha1$ and $\beta-\alpha2$), and generates the interpolated projection P_{β} as an adaptive combination of $P_{\alpha1}$ and $P_{\alpha2}$.

We propose the DRRConv and DRRTransConv, as illustrated in Fig. 3 (b), by modifying the dynamic region-aware convolution (DRConv) [28]. In the two convolutional modules, we designed: (a) the fully-connected layers that produces regulators to adapt module weights according to the angular distances between adjacent projections; and (b) region-aware convolutions that apply distinct filters to different spatial sub-regions. In contrast to DRConv, our filters are randomly initialized and not generated from the input content.

2.2.2. Deformable image registration block

The DIR block, shown in Fig. 4(a), comprises: (1) a DIR model that computes the source-to-target DVF for the full volumetric image, and (2) an intrathoracic mask module that produces a binary volume to retain only the DVF within the thoracic cavity while zeroing out values outside it.

The DIR model, shown in Fig. 4(b), adopts the VoxelMorph [29], an unsupervised learning DIR framework by mapping an input image pair to a DVF that aligns these images. The intrathoracic mask module, as illustrated in Fig. 4(c), generates a binary volume by performing slice-wise processing followed by stacking of the resulting bounding box masks. The processing workflow is as follows: (1) each slice is segmented using Otsu’s thresholding method [30] to obtain a binary foreground mask; (2) from this mask, a body binary mask is derived by identifying the largest connected component using the scikit-image library [31] in Python; (3) the lung binary mask is then obtained by subtracting the foreground mask from the body mask; and (4) the bounding box of the lung region is extracted using functions from the scikit-image toolbox [31].

2.3. Experiment

2.3.1. Data acquisition

In this experiment, we utilized 4D-CBCT projection data obtained from the publicly available 4D-Lung dataset [32] hosted by The Cancer Imaging Archive (TCIA). Note that the projections were obtained directly from the data submitter and are not available on TCIA. A total of 100 4D-CBCT datasets were acquired from 20 patients with locally advanced non-small cell lung cancer using a commercial CBCT scanner (On-Board Imager™, Varian Medical Systems, Inc.). Each 4D-CBCT dataset consisted of approximately 2500 projection images which serve as full-view projections in this work. More details about this data

¹ Note that the term “interpolation” specifically refers to the type of enhancement our model performs—compensating for insufficient imaging data to improve image quality. Conventional image enhancement, on the other hand, may also address hardware-related noise, motion artifacts, and other factors.

² The pseudocode for the overall workflow illustrated in Fig. 2 is provided in Supplementary Material S1.

³ The effect of the deformable image registration (DIR) block on image quality enhancement is described in Supplementary Material S2.

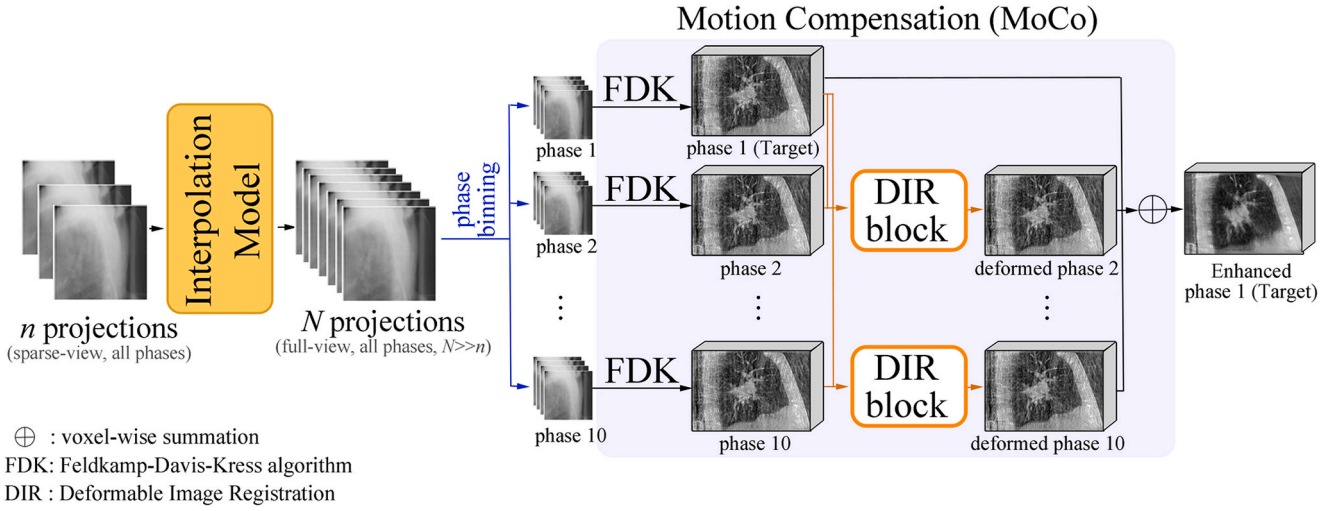


Fig. 1. The proposed Interpolation + MoCo workflow of the four-dimensional cone-beam computed tomography (4D-CBCT) reconstruction. In this figure, phase 1 is set to the target phase as an example.

can be found in Balik et al. [33] and Roman et al. [34].

All projections were down-sampled to 512×384 pixels with a pixel size of 0.776×0.776 mm. The projections were sorted into 10 respiratory phases based on the breathing signal extracted via intensity analysis [35], and reconstructed with an in-house implementation of FDK algorithm. The source-to-isocenter distance was 100 cm, and the source-to-detector distance was 150 cm. The detector was laterally offset by 150 mm. Each reconstructed volume had dimensions of $448 \times 448 \times 192$ voxels along the left-right, anterior-posterior, and superior-inferior axes, respectively, with an isotropic voxel size of 1 mm.

2.3.2. Data pre-processing for interpolation model

To simulate the sparse-view scanning for each 4D-CBCT dataset, the original set of about 2500 projections was retrospectively under-sampled to 900 projections distributed evenly over 360° . The remaining approximately 1600 projections served as ground truth for interpolation.

2.3.3. Data pre-processing for DIR model

Due to limited computational resources, all 3D-CBCT volumes were down-sampled to a size of $224 \times 224 \times 96$ voxels. Without specifying a fixed target phase, each phase in a 4D-CBCT was treated as a target and paired with one of the remaining 9 phases to form a sample pair for training the DIR model.

During testing, the output DVF of the DIR model would be up-sampled to $448 \times 448 \times 192$ voxels for motion compensation.

2.3.4. Data splitting

All data were randomly divided into a training set (12 patients), a validation set (4 patients), and a test set (4 patients), strictly based on patient identifiers. The three sets contained no overlapping patients. The splitting category for each model is detailed in Table 1.

2.3.5. Model training

All models were implemented using Pytorch, and updated with Adam [36]. The training settings were listed in Table 2. The loss function for the interpolation model is:

$$Loss_{interpolation} = 1 - SSIM(x, y) + \alpha \cdot L_{VGG}(x, y) \quad (1)$$

in which x and y refer to the interpolated projection and its ground truth respectively. L_{VGG} represents the VGG loss suggested by Ledig et al. [37]. L_{VGG} is deployed for a better high texture detail. α denotes an adjustment factor. $\alpha = 10^{-2}$. SSIM represents structural similarity index measure as

defined below:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2/2)} \quad (2)$$

where μ_x and μ_y represent the averages of x and y respectively. σ_x^2 and σ_y^2 mean the variances of x and y respectively. σ_{xy} refers to the covariance of x and y . $C_1 = (0.01 \times L)^2$. $C_2 = (0.03 \times L)^2$. L means the value range of y .

$$L_{VGG}(x, y) = MSE(\rho(x), \rho(y)) \quad (3)$$

in which MSE represents mean square error. $\rho(x)$ and $\rho(y)$ refer to the feature map output by the first nine layers of the pre-trained VGG16 [38].

The loss function for DIR model is:

$$Loss_{DIR} = 1 - SSIM(X_{tar}, \varphi(X_{src}, \phi)) + \lambda \cdot R(\phi) \quad (4)$$

in which X represents the volume reconstructed from both the sparse-view and interpolated projections. The subscripts of “tar” and “src” denote the target phase and source phase respectively. φ means the spatial transformation to warp X_{src} by the output DVF (ϕ) of DIR model. λ denotes an adjustment factor to balance SSIM and R . $\lambda = 0.01$ as suggested by Balakrishnan et al. [29]. R measures the smoothness in ϕ , which is defined as follows:

$$R(\phi) = \frac{1}{3} \left(\frac{1}{N_x} \left\| \frac{\partial \phi}{\partial x} \right\|_2^2 + \frac{1}{N_y} \left\| \frac{\partial \phi}{\partial y} \right\|_2^2 + \frac{1}{N_z} \left\| \frac{\partial \phi}{\partial z} \right\|_2^2 \right) \quad (5)$$

where N_x , N_y and N_z refer to the volume size of X_{src} along x , y and z axes. $\| \cdot \|_2$ means L2-norm.

2.3.6. Evaluation of Interpolation

The interpolated new projections were combined with the original sparse-view projections, and used to reconstruct 10-phase 3D-CBCT volumes using the FDK algorithm.

To evaluate the interpolation performance, these reconstructed volumes (denoted as X) were compared with the reconstructed volumes using full sampling projections (denoted as Y) using root mean square error (RMSE), SSIM and peak signal-to-noise ratio (PSNR). The formulas of RMSE and PSNR are as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum ((X - Y)^2 \times V_{body})} \quad (6)$$

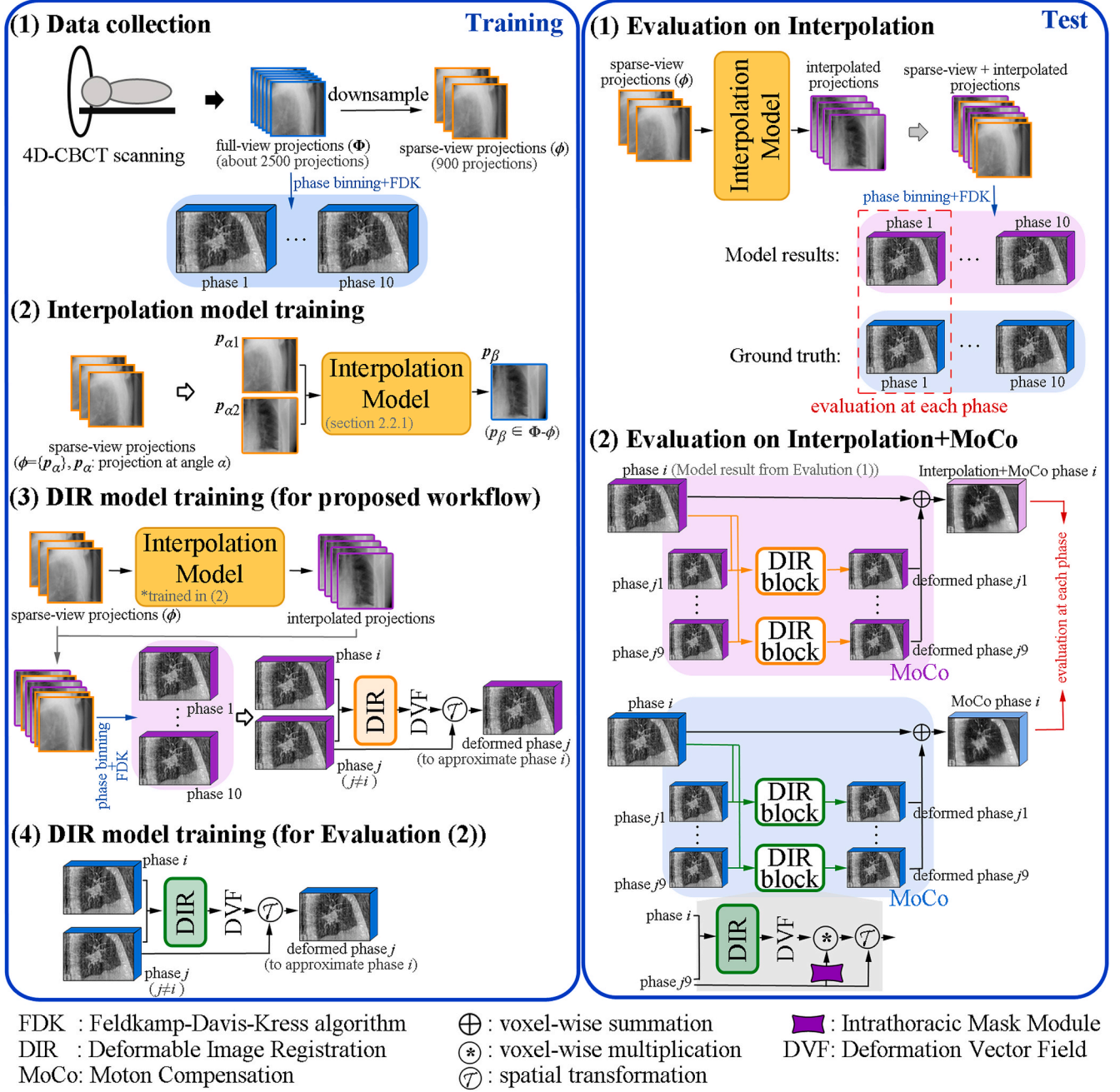


Fig. 2. Schematic representation of our Interpolation + MoCo workflow.

in which V_{body} is a binary body volume sharing the same size of X . N means the number of voxels in V_{body} whose value equals 1.

$$SSIM(X, Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2/2)} \quad (7)$$

where C_1 and C_2 have the same meaning in formula (2), while the averages of μ_X and μ_Y , the variances of σ_X^2 and σ_Y^2 , and the co-variance σ_{XY} are calculated over voxels within V_{body} whose value is 1.

$$PSNR = 20 \bullet \log_{10}(\max(Y \times V_{\text{body}})/RMSE) \quad (8)$$

where $\max(\cdot)$ refers to the maximum operation.

2.3.7. Evaluation of Interpolation + MoCo

The reconstructed volumes X and Y were further enhanced using

motion compensation method, denoted as X_{MoCo} and Y_{MoCo} respectively. They were compared using RMSE, SSIM and PSNR to assess performance.

2.3.8. Ablation study

Our key contribution is DRRConv and DRRTransConv designed to interpolate unavailable projections and reduce under-sampling artifacts in sparse-view scanning. To assess their superiority, we replaced them with conventional convolutions in the interpolation model. Their performance difference was evaluated using RMSE, SSIM and PSNR as described in Section 2.3.6, as well as outlier ratio (OR) [39,40] of local SSIM map [41] to quantify the local structural similarity.

$$OR = N_{\text{outlier}}/N_{\text{total}} \quad (9)$$

where N_{total} is the total number of voxels in V_{body} whose value is 1, and

(b) DRRConv and DRRTransConv

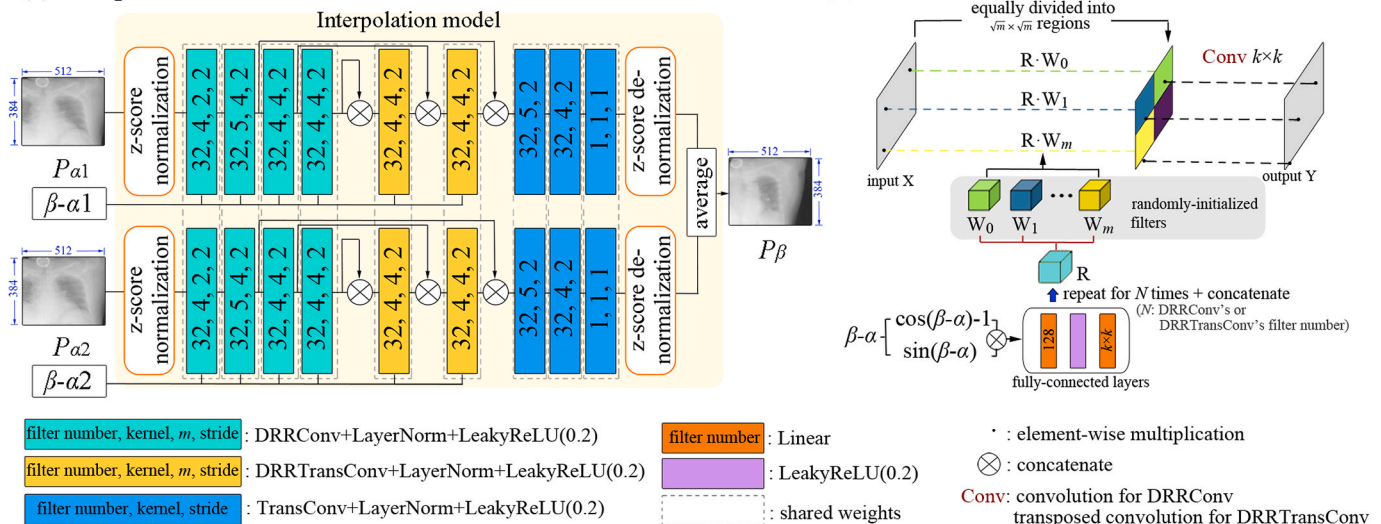


Fig. 3. Architectures of (a) the interpolation model and (b) our proposed DRRConv and DRRTransConv. Abbreviation: P_α , projection at the angle of α ; DRRConv, dynamic region- and rotation-aware convolution; DRRTransConv, dynamic region- and rotation-aware transposed convolution; TransConv, transposed convolution. (The proposed interpolation model, together with the DRRConv and DRRTransConv operations illustrated in Fig. 3, is formalized in Algorithm 3 in Supplementary Material S3.)

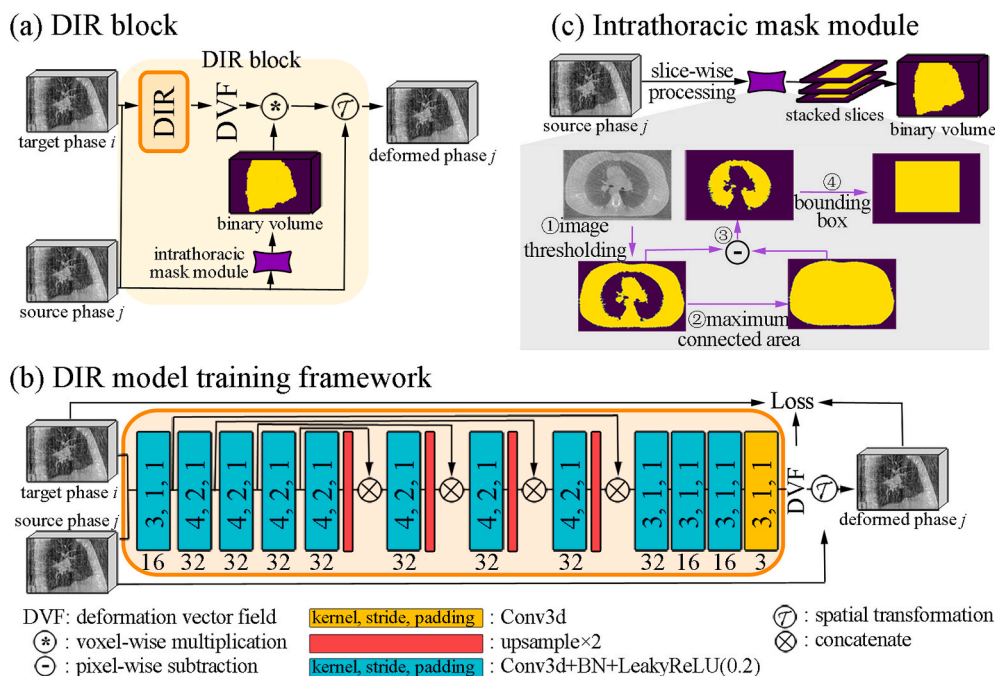


Fig. 4. Illustration of deformable image registration (DIR) block. **Abbreviations:** BN, Batch Normalization; Conv3d, convolution in three dimensions.

N_{outlier} is the number of outliers. A voxel is considered an outlier if its local SSIM score is less than 0.5. The threshold of 0.5 is determined as the average lower fence across all test cases, based on the boxplot method [42]. The local SSIM map is calculated as:

$$localSSIMmap = \{localSSIM_{ijk} | V_{body}(i, j, k) = 1\} \quad (10)$$

where the local SSIM map contains the same number of elements as the number of voxels in V_{body} with a value of 1. (i, j, k) denotes the coordinates. $localSSIM_{ijk}$ represents the structural similarity score computed between the corresponding regions of the two volumes being compared:

$$localSSIM_{ijk} = \frac{(2\mu_{x,ijk}\mu_{y,ijk} + C_1)(2\sigma_{xy,ijk} + C_2)}{(\mu_{x,ijk}^2 + \mu_{y,ijk}^2 + C_1)(\sigma_{x,ijk}^2 + \sigma_{y,ijk}^2 + C_2/2)} \quad (11)$$

in which C_1 and C_2 have the same meaning in Eq. (2), while the averages of $\mu_{x,ijk}$ and $\mu_{y,ijk}$, the variances of $\sigma_{x,ijk}^2$ and $\sigma_{y,ijk}^2$ and the co-variance of $\sigma_{xy,ijk}$ are computed within a local 7×7 window along the coronal plane using circularly symmetric Gaussian weighting. The subscripts of ‘x’ and ‘y’ denote two compared volumes.

Table 1
Data splitting details for different models.^a

Models	Input variable	Output variable	Quantity of samples ^b		
			Training	Validation	Test
Interpolation model	$P_{\alpha_1}, P_{\alpha_2}$	$P_{\beta} (\alpha_1 < \beta < \alpha_2)$	90190	30245	30084
DIR model	3D-CBCT at phases i and j	\	5400	200	200

^a Abbreviations: P_{α} , projection at the angle of α ; DIR, deformable image registration; 3D-CBCT, three-dimensional cone-beam computed tomography; \, not applicable.

^b For interpolation model, sample sets = $n \times 5$ 4D-CBCT sets per patient $\times$ the number of P_{β} per set. For DIR model, sample pairs = $n \times 5$ 4D-CBCT sets per patient $\times 10$ target-phase volumes per set. Here, n denotes the number of patients ($n = 12$ for training, 4 for validation, and 4 for test).

Table 2
Training settings for different models.^a

Models	Learning rate ^b	Batch size	Epoch	Optimal model
Interpolation model	10^{-3}	1	300	The one with a minimum loss on validation set
DIR model	10^{-3}	1	300	

^a Abbreviations: DIR, deformation image registration.

^b The learning rate is reduced by a factor of 0.1 if no loss decreases on the training set is seen for 10 epochs.

2.3.9. Comparison with other works

Our model was compared with existing methods that had been evaluated using the publicly available Varian datasets from the SPARSe-view REconstruction (SPARE) Challenge for 4D-CBCT [6]. To ensure consistency with the comparative methods, the 4D-CBCT images were reconstructed with dimensions of $450 \times 220 \times 450$ along the left-right, superior-inferior, and anterior-posterior axes, at an isotropic resolution of $1 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$. The reconstructed images were then evaluated using RMSE and normalized cross-correlation (NCC) within each of the body, lung, and planning target volume (PTV), respectively. The NCC was calculated as:

$$NCC(X, \hat{X}) = \frac{1}{M} \sum_{m=1}^M \frac{(X_m - \mu_X)(\hat{X}_m - \mu_{\hat{X}})}{\sigma_X \sigma_{\hat{X}}} \quad (12)$$

where M is the total number of voxels within the specific volume. μ and σ denote the mean and standard deviation of that volume. The comparative methods are:

- CNN enhanced [20]: A 3D convolutional neural network (CNN) applied to mitigate artifacts in 4D-CBCT images;
- Conventional MoCo [20]: Motion-compensated FDK reconstruction using DVFs estimated from initially FDK-reconstructed 4D-CBCT images;
- CNN + MoCo [20]: Motion-compensated FDK reconstruction using DVFs estimated from artifact-mitigated 4D-CBCT images generated by the CNN-based enhancement method.

3. Results

3.1. Testing results on interpolation

Table 3 summarizes the quantitative results comparing volumes reconstructed from a combination of sparse-view and interpolated projections with those reconstructed from full-view projections, demonstrating a great match. The interpolation model achieved an SSIM of 0.9657 ± 0.0155 (average $\pm$ standard deviation), an RMSE of $(8.38 \pm 0.98) \times 10^{-4} \text{ mm}^{-1}$, and a PSNR of $28.44 \pm 1.56 \text{ dB}$, substantially improving upon the sparse-view reconstructions, which yielded an SSIM of 0.8988 ± 0.0581 , an RMSE of $(16.48 \pm 4.23) \times 10^{-4} \text{ mm}^{-1}$, and a PSNR of $22.77 \pm 1.89 \text{ dB}$. All p -values < 0.05 indicate that the improvements are statistically significant.

Fig. 5 provides a comparison of three image quality metrics between volumes reconstructed using both interpolated and sparse-view projections and those reconstructed using only sparse-view projections. The differences which are highlighted in shade regions demonstrate the effectiveness of our interpolation model.

3.2. Testing results on Interpolation + MoCo

Table 4 provides a quantitative comparison between motion-compensated reconstructions from sparse and interpolated projections and those from full-view data. The proposed Interpolation + MoCo workflow significantly enhanced the reconstruction quality, achieving an SSIM of 0.9905 ± 0.0061 , an RMSE of $(32.46 \pm 5.14) \times 10^{-4} \text{ mm}^{-1}$ and a PSNR of $35.09 \pm 1.65 \text{ dB}$, compared to the sparse-view + MoCo baseline which attained an SSIM of 0.9887 ± 0.0055 , an RMSE of $(47.89 \pm 8.41) \times 10^{-4} \text{ mm}^{-1}$, and a PSNR of $31.73 \pm 1.10 \text{ dB}$. All p -values < 0.05 indicate statistically significant enhancements.

The increase in RMSE values from the “Interpolation” results in Table 3 to the “Interpolation + MoCo” results in Table 4 doesn’t indicate degraded image quality due to the addition of the “MoCo” stage, as the ground truths used for the two evaluations differ. This increase is primarily attributed to the accumulation operation involved in the motion compensation process. Further analysis is provided in the Supplementary Material S4.

Fig. 6 visually compares the three metrics between “Sparse-view +

Table 3
Quantitative results for assessing interpolation.^{a,b}

	Sparse-view SSIM	RMSE (10^{-4} mm^{-1})	PSNR (dB)	Interpolation SSIM	RMSE (10^{-4} mm^{-1})	PSNR (dB)
mean $\pm$ std	0.8988 ± 0.0581	16.48 ± 4.23	22.77 ± 1.89	0.9657 ± 0.0155	8.38 ± 0.98	28.44 ± 1.56
median	0.9199	15.58	22.98	0.9704	8.39	28.29
70 th percentile	0.9383	18.43	23.88	0.9751	8.72	29.39
90 th percentile	0.9492	21.82	24.88	0.9810	9.62	30.38
95 th percentile	0.9525	25.51	25.52	0.9821	9.87	30.99
p -value ^c	\	\	\	8.51×10^{-51}	1.68×10^{-78}	3.01×10^{-113}

^a The volumes are expressed in units of linear attenuation coefficient (mm^{-1}), with dimensions of $448 \times 448 \times 192$ voxels and an isotropic voxel size of 1 mm . The three metrics were calculated within body. “Interpolation” means the evaluated volume was reconstructed using interpolated and sparse-view projections. “Sparse-view” refers to the reconstructed volume using sparse-view projections alone.

^b Abbreviations: SSIM, structural similarity index measure; RMSE, root mean square error; PSNR, peak signal-to-noise ratio; std, standard deviation; \, not applicable.

^c The p -value is computed using a paired t -test to assess whether two paired means differ significantly. A p -value < 0.05 indicates a significant difference; otherwise, the difference is not significant.

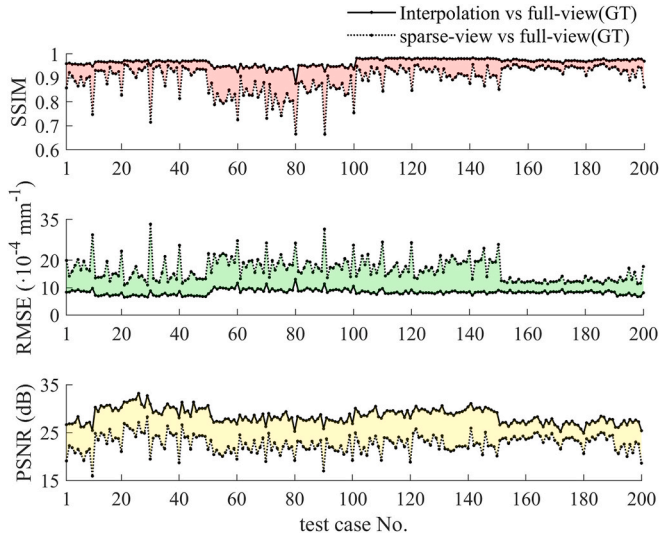


Fig. 5. Illustration of image enhancement by comparing “Interpolation” volumes with sparse-view reconstructions. Their differences are filled by the shaded regions. A higher SSIM, lower RMSE and higher PSNR correspond to a better result. All metrics are calculated within body. “GT” denotes the ground truth. “Interpolation” indicates the reconstructed volume using interpolated and sparse-view projections.

MoCo” and “Interpolation + MoCo” reconstructions, demonstrating the overall effectiveness of our proposed workflow. The two approaches yield comparable SSIM values, with “Interpolation + MoCo” performing slightly better in 144 out of the 200 test cases. The remaining 56 cases exhibited slightly lower SSIM for Interpolation + MoCo results. A detailed analysis of these cases is included in the Supplementary Material S5. The “Interpolation + MoCo” reconstruction achieved substantially lower RMSE and higher PSNR compared to “Sparse-view + MoCo” reconstruction.

Fig. 7 shows an example for qualitative evaluation. After the interpolation stage—the first step of our proposed workflow—the reconstructed volume demonstrates quality comparable to that of the full-view reconstruction. Subsequently, motion compensation—the second step of our workflow—further mitigates streak artifacts.

3.3. Ablation study

The boxplots of four evaluation metrics are shown in Fig. 8A. Our customized DRRConv and DRRTransConv achieve lower overall RMSE and lower OR compared to conventional convolutions, while maintaining comparable SSIM and PSNR. Fig. 8B presents a visual comparison of volumes reconstructed from sparse-view projections combined

with either customized- or conventional-convolution interpolated projections. The results demonstrate that our DRRConv and DRRTransConv achieve higher accuracy in reconstructed voxel values compared to conventional convolution operations.

3.4. Comparison results with other works

The comparison results were listed in Table 5, showing that our proposed approach outperforms other methods.

Note that in Table 5, although our proposed Interpolation stage presents the best performance, this does not imply that our proposed MoCo stage degrades the final image quality. It is because that the “ground truth” images defined in the SPARE challenge were reconstructed using the FDK algorithm with full projection data and still contained streak artifacts; therefore, they do not represent perfect reference images for evaluation. While MoCo improves the image quality by mitigating streak artifacts through motion compensation, it yields lower similarity metric values when compared against these imperfect “ground truth” images that themselves contain streak artifacts. This effect is evident in Fig. 7, where the subfigures labeled “full-view” and “Interpolation + MoCo” highlight the difference, with “full-view” corresponding to the ground truth in the SPARE challenge.

4. Discussion

4D-CBCT provides valuable on-site 4D volumetric imaging for motion-sensitive targets, such as lung tumors, in conventional radiotherapy as well as in emerging treatment modalities including FLASH radiotherapy [43] and proton therapy [44]. However, its intrinsically slow acquisition (8–10 min [9]) limits its clinical application and increases dose exposure to patients. To this end, numerous methods [45–47] have been proposed to accelerate 4D-CBCT acquisition while ensuring high image quality.

This study proposes an Interpolation + MoCo workflow that first interpolates missing projections from sparse-view acquisitions and then applies a motion-compensated FDK method to further address under-sampling issues caused by breathing phase sorting. The three columns under “Interpolation” in Table 3 demonstrate the effectiveness of the “Interpolation” stage, with average $\pm$ standard deviation values of 0.9657 ± 0.0155 for SSIM, $(8.38 \pm 0.98) \times 10^{-4} \text{ mm}^{-1}$ for RMSE, and $28.44 \pm 1.56 \text{ dB}$ for PSNR. The comparison of SSIM and PSNR values under the “Interpolation” column in Table 3 and “Interpolation + MoCo” column in Table 4 demonstrate that the addition of the “MoCo” stage further enhances image quality, increasing the average SSIM from 0.9657 to 0.9905 and the average PSNR from 28.44 dB to 35.09 dB. Although RMSE increases from “Interpolation” to “Interpolation + MoCo” as shown in Table 3 and 4, this does not necessarily imply degraded quality, since they are compared against different ground-

Table 4
Quantitative results for assessing Interpolation + MoCo.^{a,b}

	Sparse-view + MoCo			Interpolation + MoCo		
	SSIM	RMSE (10^{-4} mm^{-1})	PSNR (dB)	SSIM	RMSE (10^{-4} mm^{-1})	PSNR (dB)
mean $\pm$ std	0.9887 ± 0.0055	47.89 ± 8.41	31.73 ± 1.10	0.9905 ± 0.0061	32.46 ± 5.14	35.09 ± 1.65
median	0.9912	47.44	31.83	0.9932	31.86	35.84
70 th percentile	0.9926	52.39	32.37	0.9955	34.56	36.24
90 th percentile	0.9936	59.59	32.95	0.9965	40.12	36.47
95 th percentile	0.9939	61.93	33.25	0.9968	41.80	36.73
<i>p</i> -value ^c	\	\	\	2.78×10^{-4}	4.34×10^{-70}	3.18×10^{-79}

^a The volumes are expressed in units of linear attenuation coefficient (mm^{-1}), with dimensions of $448 \times 448 \times 192$ voxels and an isotropic voxel size of 1 mm. The three metrics were calculated within body. “Sparse-view + MoCo” refers to motion-compensated volumes reconstructed from sparse-view projections. “Interpolation + MoCo” denotes volumes first reconstructed from a combination of sparse-view and interpolated projections, followed by motion compensation.

^b Abbreviations: SSIM, structural similarity index measure; RMSE, root mean square error; PSNR, peak signal-to-noise ratio; std, standard deviation. \, not applicable.

^c The *p*-value is computed using a paired *t*-test to assess whether two paired means differ significantly. A *p*-value < 0.05 indicates a significant difference; otherwise, the difference is not significant.

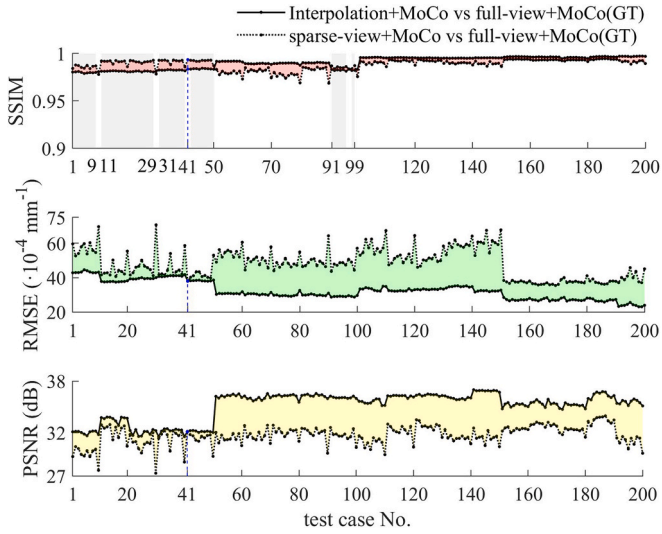


Fig. 6. Illustration of image enhancement by comparing Interpolation + MoCo volumes with sparse-view + MoCo reconstructions. Their differences are filled by the shaded regions. Gray regions mark 55 cases where Interpolation + MoCo has slightly lower SSIM, but better RMSE and PSNR compared to sparse-view + MoCo. The blue dot highlights case No.41 in which Interpolation + MoCo underperforms sparse-view + MoCo in all three metrics. A higher SSIM, lower RMSE and higher PSNR correspond to a better result. All metrics are calculated within body. “GT” denotes the ground truth. “Interpolation + MoCo”, “sparse-view + MoCo” and “full-view + MoCo” refer to motion-compensated volumes reconstructed from interpolated and sparse-view projections, sparse-view projections alone, and full-view projections, respectively.

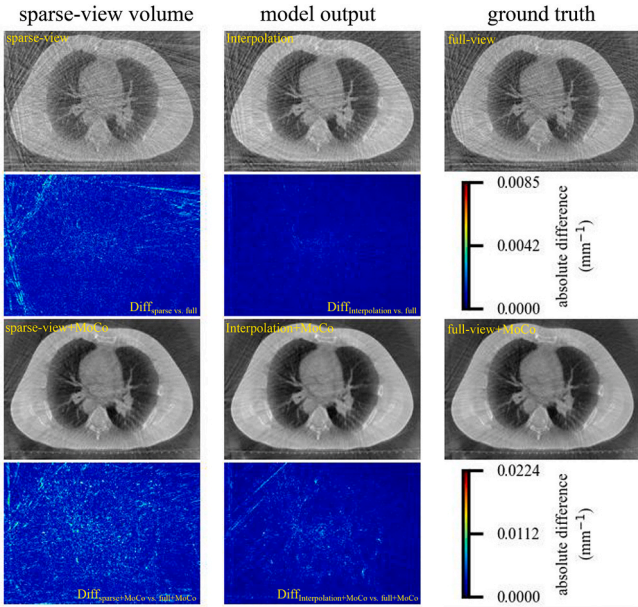


Fig. 7. Qualitative example for Interpolation and Interpolation + MoCo in transverse plane (The coronal and sagittal planes are shown in Supplementary Material S6.). Annotations: sparse-view, the reconstructed volume using sparse-view projection alone; sparse-view + MoCo, the motion-compensated “sparse-view” volume; Interpolation, the reconstructed volume using interpolated and sparse-view projections; Interpolation + MoCo, the result of applying motion compensation to the “Interpolation” volume; full-view, the reconstructed volume using full-view projections; full-view + MoCo, the volume obtained by applying motion compensation to the full-view reconstruction; Diff_{A vs. B}, absolute difference map between approaches A and B. In the subscripts of Diff, “sparse”, “full”, “sparse + MoCo” and “full + MoCo” represent “sparse-view”, “full-view”, “sparse-view + MoCo” and “full-view + MoCo” respectively.

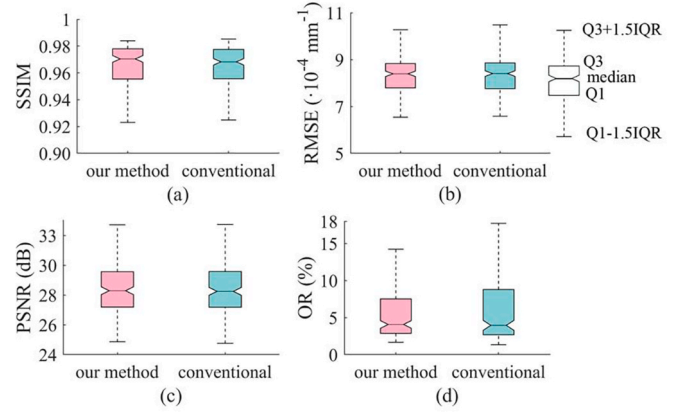


Fig. 8A. Boxplots of four evaluation metrics between our customized convolutions (DRRConv and DRRTransConv) and conventional convolutions. All metrics were calculated within body. Abbreviations: SSIM, structural similarity index measure; RMSE, root mean square error; PSNR, peak signal-to-noise ratio; OR, outlier ratio; Q1, 25th percentile; Q3, 75th percentile; IQR = Q3-Q1.

truth references. Instead, the increase reflects the effect of the accumulation operation inherent to the motion compensation process, as analyzed in the Supplementary Material S4. This is further supported by the visual comparison between the “Interpolation” and “Interpolation + MoCo” results which are exhibited in Fig. 7, where the streak artifacts are further alleviated after applying “MoCo” stage.

The proposed workflow incorporates two customized convolutional operations—DRRConv and DRRTransConv—to achieve improved interpolation performance. These operations account for (a) varying contribution ratios from two known projections due to differences in angular spacing, and (b) spatially varying correlations between sub-regions in the known projections and their counterparts in the target projection. In addition to the quantitative results listed under “Interpolation” column in Table 3, Fig. 5 provides a comparison between the “Interpolation” result and sparse-view reconstruction, showing improvements in SSIM, RMSE and PSNR after applying the interpolation. The enhanced visual clarity is further illustrated in Fig. 7. The ablation study results shown in Fig. 8A demonstrate that DRRConv and DRRTransConv outperform standard convolutions in preserving details, as evidenced by lower OR and RMSE values. Fig. 8B further illustrates the superior preservation of details, focusing on the accuracy of the reconstructed voxel values.

4.1. Design principles of customized convolutional modules

The two customized convolutional modules—DRRConv and DRRTransConv—are designed to align with rotation-induced data regularities inherent in tomographic imaging from rotational acquisitions. These regularities include:

- The contributions of neighboring projections to a target projection varies with their angular distances (as shown in Fig. 9 (a)).
- Different sub-regions within a neighboring projection exhibit distinct mappings to the target projection due to variations in their beam paths, which traverse different anatomical structures (as shown in Fig. 9 (b)).

To match regularity (a), DRRConv and DRRTransConv incorporate fully connected layers to generate regulators (denoted as R in Fig. 9(c)), which adapt the module weights dynamically. To match regularity (b), the input image is partitioned into sub-regions, each processed with distinct filter tailored to local anatomical variations.

Leveraging DRRConv and DRRTransConv, this study structurally integrates the interpolation model with intrinsic data features, achieving

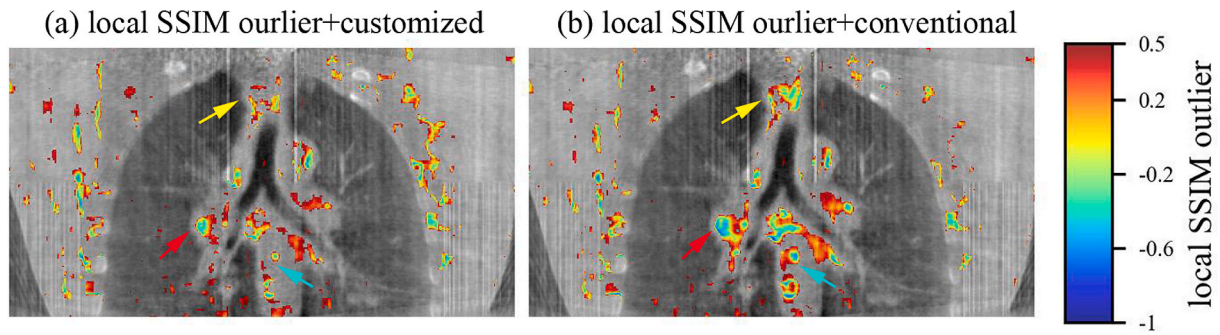


Fig. 8B. Local SSIM outliers on reconstructed volumes using (a) customized and (b) conventional convolutions. Worse results occur as local SSIM approaches -1, with outliers defined by a threshold of 0.5. Yellow, blue, and red arrows indicate areas where the customized convolutions demonstrate superior performance. Annotations: customized, interpolation model employs customized convolutions; conventional, interpolation model uses conventional convolutions; SSIM, structural similarity index measure.

Table 5

Comparison results using the Varian datasets from the SPARE challenge.^a

Case No.	Model	Body RMSE	NCC	Lung RMSE	NCC	PTV RMSE	NCC
1	CNN enhanced	2.7	0.93	2.3	0.76	2.7	0.90
	Conventional	1.9	0.96	1.8	0.83	1.9	0.93
	MoCo						
	CNN + MoCo	1.8	0.96	1.8	0.85	2.0	0.93
	Interpolation	1.7	0.97	1.2	0.93	1.2	0.95
2	Interpolation + MoCo	2.0	0.95	1.6	0.88	1.5	0.91
	CNN enhanced	2.9	0.96	3.9	0.85	2.4	0.76
	Conventional	2.5	0.93	2.7	0.76	3.2	0.58
	MoCo						
	CNN + MoCo	1.8	0.96	1.9	0.87	2.2	0.76
3	Interpolation	1.4	0.98	1.4	0.93	1.5	0.86
	Interpolation + MoCo	1.5	0.97	1.5	0.92	1.7	0.83
	CNN enhanced	2.9	0.94	3.4	0.77	4.0	0.75
	Conventional	2.6	0.93	2.6	0.76	3.2	0.81
	MoCo						
	CNN + MoCo	2.3	0.95	2.4	0.81	2.7	0.83
	Interpolation	2.3	0.95	2.1	0.91	2.3	0.86
	Interpolation + MoCo	2.2	0.96	2.2	0.89	2.8	0.80

^a The best results (lowest RMSE, highest NCC) for each case are in bold. The metrics were calculated in different regions. The unit of RMSE is $\times 10^{-3} \text{ mm}^{-1}$. Abbreviations: PTV, planning target volume; NCC, normalized cross-correlation.

high efficiency and accuracy via a lightweight architecture. While deeper neural networks remain popular for their performance on complex tasks, lightweight models are more favorable for widespread clinical applications. Notably, DRRConv and DRRTransConv are generalizable and plug-and-play components that extend beyond 4D-CBCT and are applicable to a wide range of projection-based imaging modalities involving rotational acquisition.

4.2. Limitation and future work

The proposed workflow was validated on a relatively small patient cohort under a specific acquisition protocol. Although the results are promising, further validation on larger and more diverse datasets is necessary to evaluate the generalizability and clinical robustness of the method. A limitation of the model is that it was trained and tested on a single acquisition protocol. Training the model on data from multiple scanners could enable it to capture more generalizable rotation-induced data patterns, thereby facilitating the application of this work across different scanning protocols using a single set of fixed model weights. In addition, the current model does not explicitly address hardware-related artifacts such as detector non-uniformity, which may require further incorporation of physics-based corrections. It is also worth noting that

the data used in this study were not flat-field corrected, which could contribute to residual artifacts.

The proposed method can be extended to more challenging tasks, such as high-sparse or ultra-sparse CBCT enhancement. These tasks involve larger angular gaps between the target and source projections, requiring a larger receptive field to capture sufficient contextual information for accurate interpolation. In addition to dilated convolutions [48], which provide large receptive fields with efficient parameter usage, large kernel adapters [49–52] also offer promising potential. In future work, we plan to integrate these approaches with our region- and rotation-aware, adaptively-tuned DRRConv and DRRTransConv modules to further improve image enhancement performance.

5. Conclusion

We proposed an Interpolation + MoCo workflow to enhance the quality of sparse-view 4D-CBCT images, as demonstrated in the experimental results using real patient data. In the “Interpolation” stage, our customized DRRConv and DRRTransConv modules interpolate missing projections from two known views by accounting for their varying contribution ratios, which result from differences in spatial sub-regions and angular spacing. Our interpolation model improved the results of reconstruction with and without motion compensation compared to sparse-view reconstruction. Ablation study demonstrated the efficacy of customized convolutional module compared to the convention module. Compared to conventional full-view 4D-CBCT imaging, which typically requires long acquisition time (8–10 min) and high imaging dose (effective dose⁴: $7.3 \pm 0.9 \text{ mSv}$ for lung [53]), the proposed workflow reduces the scan time to just 1 minute and 2.63 mSv for lung⁵ while preserving anatomical details and maintaining image quality, which greatly enhance the efficiency of 4D-CBCT for clinical use. The developed model can also be applied to enhance the quality of other imaging modalities with sparse acquisition to accelerate the scanning process.

⁴ Effective dose [54] is a dose quantity defined in the International Commission on Radiological Protection (ICRP) system of radiological protection, associated with the stochastic health risks to the whole body, such as the probability of cancer induction and hereditary effects. Effective dose for medical purposes (e.g., 4D-CBCT scanning) is not currently incorporated into radiation protection standards. However, considering the potential benefits from reduced imaging dose in this work, it is worth noting that the ICRP recommends an effective dose limit for public exposure of 1 mSv per year. In special circumstances, a single-year exposure of up to 5 mSv is permitted, provided that the 5-year average effective dose does not exceed 1 mSv per year [54].

⁵ This work reduces the number of projections from approximately 2500 (full-view scan) to 900. Assuming a linear relationship between dose and the number of projections, the imaging dose is estimated to be reduced to 2.63mSv (i.e., $7.3\text{mSv} \times 900/2500$).

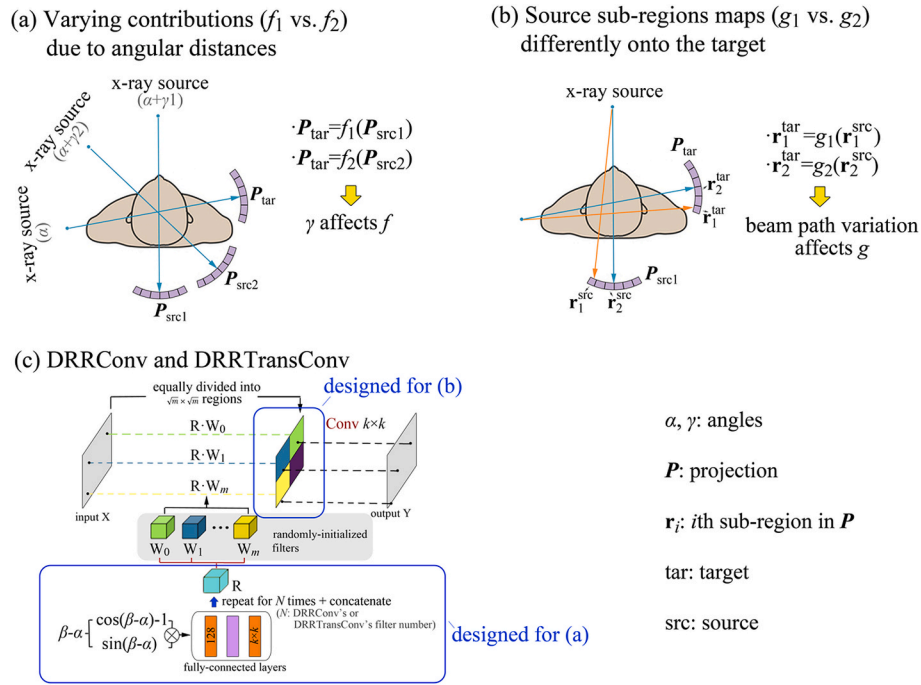


Fig. 9. Illustrations of (a-b) rotation-induced data regularities and (c) DRRConv and DRRTransConv structures, with design highlights from (a-b). (a-b) depict a one-dimensional detector for simplicity. Abbreviations in (c) are consistent with Fig. 3.

CRedit authorship contribution statement

Jie Zhang: Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Lei Ren:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study uses only publicly available datasets that were released with appropriate ethical approval and consent by their original authors. No new data involving human or animal subjects were collected or generated by the authors. All data were obtained either from open-access repositories or directly from the original authors to supplement missing components, and we strictly adhere to the terms of use and privacy policies associated with these datasets. Therefore, no institutional ethical approval was required for this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bspc.2026.109935>.

Data availability

Data will be made available on request.

References

- [1] J. Sonke, L. Zijp, P. Remeijer, M. Van Herk, Respiratory correlated cone beam CT, *Med. Phys.* 32 (2005) 1176–1186, <https://doi.org/10.1118/1.1869074>.
- [2] K. Nakagawa, A. Haga, S. Kida, Y. Masutani, H. Yamashita, W. Takahashi, A. Sakumi, N. Saotome, T. Shiraki, K. Ohtomo, Y. Iwai, K. Yoda, 4D registration and 4D verification of lung tumor position for stereotactic volumetric modulated arc therapy using respiratory-correlated cone-beam CT, *J. Radiat. Res. (Tokyo)* 54 (2013) 152–156, <https://doi.org/10.1093/jrr/rrs058>.
- [3] R.A. Sweeney, B. Seubert, S. Stark, V. Homann, G. Müller, M. Flentje, M. Guckenberger, Accuracy and inter-observer variability of 3D versus 4D cone-beam CT based image-guidance in SBRT for lung tumors, *Radiat. Oncol.* 7 (2012) 81, <https://doi.org/10.1186/1748-717X-7-81>.
- [4] Y. Zhang, Z. Jiang, Y. Zhang, L. Ren, A review on 4D cone-beam CT (4D-CBCT) in radiation therapy: technical advances and clinical applications, *Med. Phys.* 51 (2024) 5164–5180, <https://doi.org/10.1002/mp.17269>.
- [5] Z. Dong, S. Yu, A. Szmul, J. Wang, J. Qi, H. Wu, J. Li, Z. Lu, Y. Zhang, Simulation of a new respiratory phase sorting method for 4D-imaging using optical surface information towards precision radiotherapy, *Comput. Biol. Med.* 162 (2023) 107073, <https://doi.org/10.1016/j.combiomed.2023.107073>.
- [6] C. Shieh, Y. Gonzalez, B. Li, X. Jia, S. Rit, C. Mory, M. Riblett, G. Hugo, Y. Zhang, Z. Jiang, X. Liu, L. Ren, P. Keall, SPARE: sparse-view reconstruction challenge for 4D cone-beam CT from a 1-min scan, *Med. Phys.* 46 (2019) 3799–3811, <https://doi.org/10.1002/mp.13687>.
- [7] S. Thengumpallil, K. Smith, P. Monnin, J. Bourhis, F. Bochud, R. Moeckli, Difference in performance between 3D and 4D CBCT for lung imaging: a dose and image quality analysis, *J. Appl. Clin. Med. Phys.* 17 (2016) 97–106, <https://doi.org/10.1120/jacmp.v17i6.6459>.
- [8] B.K.F. Lau, T. Reynolds, A. Wallis, S. Smith, A. George, P.J. Keall, J.-J. Sonke, S. K. Vinod, O. Dillon, R.T. O'Brien, Reducing 4DCBCT scan time and dose through motion compensated acquisition and reconstruction, *Phys. Med. Biol.* 66 (2021) 075002, <https://doi.org/10.1088/1361-6560/abebfb>.
- [9] G.D. Hugo, E. Weiss, W.C. Sleeman, S. Balik, P.J. Keall, J. Lu, J.F. Williamson, A longitudinal four-dimensional computed tomography and cone beam computed tomography dataset for image-guided radiation therapy research in lung cancer, *Med. Phys.* 44 (2017) 762–771, <https://doi.org/10.1002/mp.12059>.
- [10] X. Zhao, Y. Du, H. Yue, R. Wang, S. Zhou, H. Wu, W. Wang, Y. Peng, Deep learning-based projection synthesis for low-dose cone-beam computed tomography imaging in image-guided radiotherapy, *Quant. Imaging Med. Surg.* 14 (2024) 231–250, <https://doi.org/10.21037/qims-23-759>.
- [11] L. Zhu, Y. Han, X. Xi, H. Fu, S. Tan, M. Liu, S. Yang, C. Liu, L. Li, B. Yan, STEDNet: Swin transformer-based encoder–decoder network for noise reduction in low-dose CT, *Med. Phys.* 50 (2023) 4443–4458, <https://doi.org/10.1002/mp.16249>.

- [12] M. Jian, X. Yu, H. Zhang, C. Yang, SwinCT: feature enhancement based low-dose CT images denoising with swin transformer, *Multimed. Syst.* 30 (2024) 1, <https://doi.org/10.1007/s00530-023-01202-x>.
- [13] L. Chao, Y. Wang, T. Zhang, W. Shan, H. Zhang, Z. Wang, Q. Li, Joint denoising and interpolating network for low-dose cone-beam CT reconstruction under hybrid dose-reduction strategy, *Comput. Biol. Med.* 168 (2024) 107830, <https://doi.org/10.1016/j.combiomed.2023.107830>.
- [14] Y. Wang, L. Chao, W. Shan, H. Zhang, Z. Wang, Q. Li, Improving the Quality of Sparse-view Cone-Beam Computed Tomography via Reconstruction-Friendly Interpolation Network, in: L. Wang, J. Gall, T.-J. Chin, I. Sato, R. Chellappa (Eds.), *Comput. Vis. – ACCV 2022*, Springer Nature Switzerland, Cham, 2023: pp. 86–100. https://doi.org/10.1007/978-3-031-26351-4_6.
- [15] K. Lu, Z. Zhang, L. Ren, F. Yin, Deep residual network-based projection interpolation and post-processing techniques for thoracic patient CBCT reconstruction, *Med. Phys.* 52 (2025) e17953, <https://doi.org/10.1002/mp.17953>.
- [16] L.A. Feldkamp, L.C. Davis, J.W. Kress, Practical cone-beam algorithm, *J. Opt. Soc. Am. A* 1 (1984) 612, <https://doi.org/10.1364/JOSAA.1.000612>.
- [17] Z. Jiang, Y. Chang, Z. Zhang, F. Yin, L. Ren, Fast four-dimensional cone-beam computed tomography reconstruction using deformable convolutional networks, *Med. Phys.* 49 (2022) 6461–6476, <https://doi.org/10.1002/mp.15806>.
- [18] Z. Jiang, Z. Zhang, Y. Chang, Y. Ge, F.-F. Yin, L. Ren, Enhancement of 4-D cone-beam computed tomography (4D-CBCT) using a dual-encoder convolutional neural network (DeCNN), *IEEE Trans. Radiat. Plasma Med. Sci.* 6 (2022) 222–230, <https://doi.org/10.1109/TRPMS.2021.3133510>.
- [19] Z. Jiang, Z. Zhang, Y. Chang, Y. Ge, F.-F. Yin, L. Ren, Prior image-guided cone-beam computed tomography augmentation from under-sampled projections using a convolutional neural network, *Quant. Imaging Med. Surg.* 11 (2021) 4767–4780. <https://doi.org/10.21037/qims-21-114>.
- [20] Z. Zhang, J. Liu, D. Yang, U.S. Kamilov, G.D. Hugo, Deep learning-based motion compensation for four-dimensional cone-beam computed tomography (4D-CBCT) reconstruction, *Med. Phys.* 50 (2023) 808–820, <https://doi.org/10.1002/mp.16103>.
- [21] Z. Zhang, Y. Hao, X. Jin, D. Yang, U.S. Kamilov, G.D. Hugo, Fast motion-compensated reconstruction for 4D-CBCT using deep learning-based groupwise registration, *Biomed. Phys. Eng. Express* 11 (2025) 015030, <https://doi.org/10.1088/2057-1976/ad97c1>.
- [22] Z. Jiang, Y. Chen, Y. Zhang, Y. Ge, F.-F. Yin, L. Ren, Augmentation of CBCT reconstructed from under-sampled projections using deep learning, *IEEE Trans. Med. Imaging* 38 (2019) 2705–2715, <https://doi.org/10.1109/TMI.2019.2912791>.
- [23] X. Zhao, Y. Du, Y. Peng, DLPVI: Deep learning framework integrating projection, view-by-view backprojection, and image domains for high- and ultra-sparse-view CBCT reconstruction, *Comput. Med. Imaging Graph.* 121 (2025) 102508, <https://doi.org/10.1016/j.compmedimag.2025.102508>.
- [24] S. Rit, J.W.H. Wolthaus, M. Van Herk, J. Sonke, On-the-fly motion-compensated cone-beam CT using an a priori model of the respiratory motion, *Med. Phys.* 36 (2009) 2283–2296, <https://doi.org/10.1118/1.3115691>.
- [25] O. Dillon, P.J. Keall, C.-C. Shieh, R.T. O'Brien, Evaluating reconstruction algorithms for respiratory motion guided acquisition, *Phys. Med. Biol.* 65 (2020) 175009, <https://doi.org/10.1088/1361-6560/ab98d3>.
- [26] O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional Networks for Biomedical Image Segmentation, (2015). <https://doi.org/10.48550/ARXIV.1505.04597>.
- [27] J.L. Ba, J.R. Kiros, G.E. Hinton, Layer Normalization, (2016). <https://doi.org/10.48550/ARXIV.1607.06450>.
- [28] J. Chen, X. Wang, Z. Guo, X. Zhang, J. Sun, Dynamic Region-Aware Convolution (2021). <https://doi.org/10.48550/arXiv.2003.12243>.
- [29] G. Balakrishnan, A. Zhao, M.R. Sabuncu, J. Guttag, A.V. Dalca, VoxelMorph: a learning framework for deformable medical image registration, *IEEE Trans. Med. Imaging* 38 (2019) 1788–1800, <https://doi.org/10.1109/TMI.2019.2897538>.
- [30] N. Otsu, A threshold selection method from gray-level histograms, *IEEE Trans. Syst. Man Cybern.* 9 (1979) 62–66, <https://doi.org/10.1109/TSMC.1979.4310076>.
- [31] S. Van Der Walt, J.L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J.D. Warner, N. Yager, E. Gouillart, T. Yu, scikit-image: image processing in Python, *PeerJ* 2 (2014) e453.
- [32] G.D. Hugo, E. Weiss, W.C. Sleeman, S. Balik, P.J. Keall, J. Lu, J.F. Williamson, Data from 4D Lung Imaging of NSCLC Patients, (2016). <https://doi.org/10.7937/K9/TCIA.2016.ELN8YGLE>.
- [33] S. Balik, E. Weiss, N. Jan, N. Roman, W.C. Sleeman, M. Fatyga, G.E. Christensen, C. Zhang, M.J. Murphy, J. Lu, P. Keall, J.F. Williamson, G.D. Hugo, Evaluation of four-dimensional computed tomography to four-dimensional cone-beam computed tomography deformable image registration for lung cancer adaptive radiation therapy, *Int. J. Radiat. Oncol.* 86 (2013) 372–379, <https://doi.org/10.1016/j.ijrobp.2012.12.023>.
- [34] N.O. Roman, W. Shepherd, N. Mukhopadhyay, G.D. Hugo, E. Weiss, Interfractional positional variability of fiducial markers and primary tumors in locally advanced non-small-cell lung cancer during audiovisual biofeedback radiotherapy, *Int. J. Radiat. Oncol.* 83 (2012) 1566–1572, <https://doi.org/10.1016/j.ijrobp.2011.10.051>.
- [35] A. Kavanagh, P.M. Evans, V.N. Hansen, S. Webb, Obtaining breathing patterns from any sequential thoracic x-ray image set, *Phys. Med. Biol.* 54 (2009) 4879–4888, <https://doi.org/10.1088/0031-9155/54/16/003>.
- [36] D.P. Kingma, J. Ba, Adam: A Method for Stochastic Optimization, (2017). <https://doi.org/10.48550/arXiv.1412.6980>.
- [37] C. Ledig, L. Theis, F. Huszar, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang, W. Shi, Photo-realistic single image super-resolution using a generative adversarial network, (2016). <https://doi.org/10.48550/ARXIV.1609.04802>.
- [38] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, (2014). <https://doi.org/10.48550/ARXIV.1409.1556>.
- [39] D.M. Chandler, Most apparent distortion: full-reference image quality assessment and the role of strategy, *J. Electron. Imaging* 19 (2010) 011006, <https://doi.org/10.1117/1.3267105>.
- [40] H.R. Sheikh, M.F. Sabir, A.C. Bovik, A statistical evaluation of recent full reference image quality assessment algorithms, *IEEE Trans. Image Process.* 15 (2006) 3440–3451, <https://doi.org/10.1109/TIP.2006.881959>.
- [41] A.C. Zhou Wang, H.R. Bovik, E.P.S. Sheikh, Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* 13 (2004) 600–612, <https://doi.org/10.1109/TIP.2003.819861>.
- [42] S.H.C. Du Toit, A.G.W. Steyn, R.H. Stumpf, Graphical exploratory data analysis, Springer New York, New York, NY (1986), <https://doi.org/10.1007/978-1-4612-4950-4>.
- [43] I. El Naqa, B.W. Pogue, R. Zhang, I. Oraiqat, K. Parodi, Image guidance for FLASH radiotherapy, *Med. Phys.* 49 (2022) 4109–4122, <https://doi.org/10.1002/mp.15662>.
- [44] T. Jenny, A. Duetschler, A. Giger, O. Pusterla, S. Safai, D.C. Weber, A.J. Lomax, Y. Zhang, Technical note: towards more realistic 4DCT(MRI) numerical lung phantoms, *Med. Phys.* 51 (2024) 579–590, <https://doi.org/10.1002/mp.16451>.
- [45] S.A. Yoganathan, K.J. Maria Das, S. Mohamed Ali, A. Agarwal, S.P. Mishra, S. Kumar, Evaluating the four-dimensional cone beam computed tomography with varying gantry rotation speed, *Br. J. Radiol.* 89 (2016) 20150870, <https://doi.org/10.1259/bjr.20150870>.
- [46] X. Zhao, Y. Du, Y. Peng, Deep learning-based multi-view projection synthesis approach for improving the quality of sparse-view CBCT in image-guided radiotherapy, *J. Imaging Inform. Med.* (2025), <https://doi.org/10.1007/s10278-025-01390-0>.
- [47] D. Hu, Y. Zhang, J. Liu, Y. Zhang, J.L. Coatrieux, Y. Chen, PRIOR: prior-regularized iterative optimization reconstruction For 4D CBCT, *IEEE J. Biomed. Health Inform.* 26 (2022) 5551–5562, <https://doi.org/10.1109/JBHI.2022.3201232>.
- [48] F. Yu, V. Koltun, Multi-Scale Context Aggregation by Dilated Convolutions, (2015). <https://doi.org/10.48550/ARXIV.1511.07122>.
- [49] Z. Zhu, L. Liu, R.C. Free, A. Anjum, J. Panneerselvam, OPT-CO: Optimizing pre-trained transformer models for efficient COVID-19 classification with stochastic configuration networks, *Inf. Sci.* 680 (2024) 121141, <https://doi.org/10.1016/j.ins.2024.121141>.
- [50] H. Zhu, Z. Zhu, S.-Y. Lu, BccT: an efficient transformer model for blood cell classification, *Multimed. Syst.* 32 (2026) 29, <https://doi.org/10.1007/s00530-025-02085-w>.
- [51] S.-Y. Lu, Z. Zhu, Y.-D. Zhang, Y.-D. Yao, Tuberculosis and pneumonia diagnosis in chest X-rays by large adaptive filter and aligning normalized network with report-guided multi-level alignment, *Eng. Appl. Artif. Intell.* 158 (2025) 111575, <https://doi.org/10.1016/j.engappai.2025.111575>.
- [52] Z. Zhu, S.-Y. Lu, T. Huang, L. Liu, Z. Liu, LKA: Large Kernel Adapter for Enhanced Medical Image Classification, in: J.C. Gee, D.C. Alexander, J. Hong, J.E. Iglesias, C. H. Sudre, A. Venkataraman, P. Golland, J.H. Kim, J. Park (Eds.), *Med. Image Comput. Comput. Assist. Interv. – MICCAI 2025*, Springer Nature Switzerland, Cham, 2026: pp. 394–404. https://doi.org/10.1007/978-3-032-04978-0_38.
- [53] Y. Yuasa, T. Shinoki, R. Onizuka, K. Fujimoto, Estimation of effective imaging dose and excess absolute risk of secondary cancer incidence for four-dimensional cone-beam computed tomography acquisition, *J. Appl. Clin. Med. Phys.* 20 (2019) 57–68, <https://doi.org/10.1002/acm2.12741>.
- [54] International Commission on Radiological Protection (ICRP), The 2007 Recommendations of the International Commission on Radiological Protection, *Ann. ICRP* 37 (2007) 2–4.